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*
*          STAAD.Pro
*      Version  2007    Build 04
*      Proprietary Program of
*      Research Engineers, Intl.
*      Date=    DEC 18, 2012
*      Time=    19:36:36
*
*      USER ID: Parsons Brinckerhoff
*****
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1. STAAD SPACE
INPUT FILE: 121206 Crane Staircase R1.STD
2. START JOB INFORMATION
3. ENGINEER DATE 26-NOV-12
4. JOB NAME CRANE STAIRCASE
5. END JOB INFORMATION
6. INPUT WIDTH 79
7. UNIT METER KN
8. JOINT COORDINATES
9. 1 -109.835 0 -23.25; 2 -109.835 0 -22.05; 3 -107.085 0 -23.25
10. 4 -107.085 0 -22.05; 5 -107.085 3 -23.25; 6 -107.085 3 -22.05
11. 7 -106.085 0 -23.25; 8 -106.085 0 -22.05; 9 -106.085 3 -23.25
12. 10 -106.085 3 -22.05; 11 -103.085 6.2 -23.25; 12 -103.085 6.2 -22.05
13. 13 -102.085 0 -23.25; 14 -102.085 0 -22.05; 15 -102.085 3 -23.25
14. 16 -102.085 3 -22.05; 17 -102.085 6.2 -23.25; 18 -102.085 6.2 -22.05
15. 19 -100.885 0 -23.25; 20 -100.885 0 -22.05; 21 -100.885 3 -23.25
16. 22 -100.885 3 -22.05; 23 -100.885 6.2 -23.25; 24 -100.885 6.2 -22.05
17. MEMBER INCIDENCES
18. 1 5 1; 2 6 2; 3 5 3; 4 6 4; 5 6 5; 6 9 3; 7 7 5; 8 8 6; 9 10 4; 10 9 5
19. 11 10 6; 12 9 7; 13 10 8; 14 10 9; 15 11 9; 16 12 10; 17 12 11; 18 11 17
20. 19 12 18; 20 15 13; 21 16 14; 22 16 15; 23 17 15; 24 18 16; 25 18 17; 26 19 15
21. 27 21 13; 28 20 16; 29 22 14; 30 15 21; 31 16 22; 32 17 23; 33 18 24; 34 21 19
22. 35 22 20; 36 22 21; 37 23 21; 38 24 22; 39 24 23
23. DEFINE MATERIAL START
24. ISOTROPIC STEEL
25. E 2.05E+008
26. POISSON 0.3
27. DENSITY 76.8195
28. ALPHA 1.2E-005
29. DAMP 0.03
30. END DEFINE MATERIAL
31. MEMBER PROPERTY JAPANESE
32. 20 21 23 24 34 35 37 38 TABLE ST H250X250X9
33. 3 4 12 13 TABLE ST H250X250X9
34. 5 10 11 14 22 25 30 33 36 39 TABLE ST H200X200X8
35. 1 2 15 16 19 TABLE ST C250X90X9
36. 6 7 9 26 29 TABLE ST C150X75X6.5
37. CONSTANTS
38. MATERIAL STEEL ALL
39. BETA 180 MEMB 2 16
40. BETA 90 MEMB 3 4 12 13 20 21 23 24 34 35 37 38
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41. SUPPORTS
42. 1 TO 4 7 8 13 14 20 PINNED
43. 19 FIXED BUT MY MZ KMX 1E-005
44. MEMBER RELEASE
45. 15 16 END MY MZ
46. 1 2 START MY MZ
47. 17 START MY MZ
48. 17 END MY MZ
49. 33 START MY MZ
50. 33 END MY MZ
51. 32 START MY MZ
52. 32 END MY MZ
53. 10 START MY MZ
54. 10 END MY MZ
55. 31 END MY MZ
56. 31 START MY MZ
57. 30 START MY MZ
58. 30 END MY MZ
59. 11 START MY MZ
60. 11 END MY MZ
61. LOAD 1 LOADTYPE NONE TITLE DL
62. SELFWEIGHT Y -1 LIST 1 TO 39
63. MEMBER LOAD
64. *0.75KPA DEAD LOAD*
65. 18 19 32 33 UNI GY -0.45
66. 1 2 10 11 15 16 UNI GY -0.375
67. LOAD 2 LOADTYPE NONE TITLE LL
68. *5KPA LIVE LOAD*
69. MEMBER LOAD
70. 18 19 32 33 UNI GY -3
71. 1 2 10 11 15 16 UNI GY -2.5
72. LOAD 3 LOADTYPE NONE TITLE WLX
73. MEMBER LOAD
74. 3 TO 5 12 TO 14 17 20 TO 25 34 TO 39 UNI GX -0.24
75. LOAD 4 LOADTYPE NONE TITLE WLZ
76. MEMBER LOAD
77. 1 TO 4 10 TO 13 15 16 18 TO 21 23 24 30 TO 35 37 38 UNI GZ -0.24
78. 6 TO 9 26 TO 29 UNI GY -0.15
79. LOAD 5 LOADTYPE NONE TITLE NLX
80. SELFWEIGHT X -0.015 LIST 1 TO 39
81. MEMBER LOAD
82. 18 19 32 33 UNI GX -0.007
83. 1 2 10 11 15 16 UNI GX -0.006
84. LOAD 6 LOADTYPE NONE TITLE NLZ
85. SELFWEIGHT Z -0.015 LIST 1 TO 39
86. MEMBER LOAD
87. 18 19 32 33 UNI GZ -0.007
88. 1 2 10 11 15 16 UNI GZ -0.006
89. LOAD 7 LOADTYPE NONE TITLE 1.0DL+1.0LL
90. REPEAT LOAD
91. 1 1.0 2 1.0
92. LOAD 8 LOADTYPE NONE TITLE 1.0DL+1.0LL+1.0WLX
93. REPEAT LOAD
94. 1 1.0 2 1.0 3 1.0
95. LOAD 9 LOADTYPE NONE TITLE 1.0DL+1.0LL+1.0WLZ
96. REPEAT LOAD
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97. 1 1.0 2 1.0 4 1.0
98. LOAD 10 LOADTYPE NONE TITLE 1.0DL+1.0LL+1.0NLX
99. REPEAT LOAD
100. 1 1.0 2 1.0 5 1.0
101. LOAD 11 LOADTYPE NONE TITLE 1.0DL+1.0LL+1.0NLZ
102. REPEAT LOAD
103. 1 1.0 2 1.0 6 1.0
104. LOAD 21 LOADTYPE NONE TITLE 1.0DL+1.0LL-1.0 WLX
105. REPEAT LOAD
106. 1 1.0 2 1.0 3 -1.0
107. LOAD 22 LOADTYPE NONE TITLE 1.0DL+1.0LL-1.0 WLZ
108. REPEAT LOAD
109. 4 -1.0 1 1.0 2 1.0
110. LOAD 23 LOADTYPE NONE TITLE 1.0DL+1.0LL-1.0 NLX
111. REPEAT LOAD
112. 1 1.0 2 1.0 5 -1.0
113. LOAD 24 LOADTYPE NONE TITLE 1.0DL+1.0LL-1.0 NLZ
114. REPEAT LOAD
115. 6 -1.0 1 1.0 2 1.0
116. *ULTIMATE LOAD COMBINATIONS
117. LOAD 25 LOADTYPE NONE TITLE 1.0DL+1.6LL
118. REPEAT LOAD
119. 1 1.0 2 1.6
120. LOAD 26 LOADTYPE NONE TITLE 1.4DL+1.6LL
121. REPEAT LOAD
122. 1 1.4 2 1.6
123. LOAD 27 LOADTYPE NONE TITLE 1.2DL+1.2LL+1.2WLX
124. REPEAT LOAD
125. 1 1.2 2 1.2 3 1.2
126. LOAD 28 LOADTYPE NONE TITLE 1.2DL+1.2LL+1.2WLZ
127. REPEAT LOAD
128. 1 1.2 2 1.2 4 1.2
129. LOAD 29 LOADTYPE NONE TITLE 1.2DL+1.2LL+1.2NLX
130. REPEAT LOAD
131. 1 1.2 2 1.2 5 1.2
132. LOAD 30 LOADTYPE NONE TITLE 1.2LL+1.2LL+1.2NLZ
133. REPEAT LOAD
134. 1 1.2 2 1.2 6 1.2
135. LOAD 41 LOADTYPE NONE TITLE 1.2DL+1.2LL-1.2 WLX
136. REPEAT LOAD
137. 1 1.2 2 1.2 3 -1.2
138. LOAD 42 LOADTYPE NONE TITLE 1.2DL+1.2LL-1.2WLZ
139. REPEAT LOAD
140. 4 -1.2 1 1.2 2 1.2
141. LOAD 43 LOADTYPE NONE TITLE 1.2DL+1.2LL-1.2 NLX
142. REPEAT LOAD
143. 1 1.2 2 1.2 5 -1.2
144. LOAD 44 LOADTYPE NONE TITLE 1.2DL+1.2LL-1.2 NLZ
145. REPEAT LOAD
146. 6 -1.2 1 1.2 2 1.2
147. PERFORM ANALYSIS

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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 24/ 39/ 10

SOLVER USED IS THE IN-CORE ADVANCED SOLVER

TOTAL PRIMARY LOAD CASES = 25, TOTAL DEGREES OF FREEDOM = 114

148. DEFINE ENVELOP
149. 25 TO 30 41 TO 44 ENVELOP 1 TYPE STRESS
150. 7 TO 11 21 TO 24 ENVELOP 2 TYPE SERVICEABILITY
151. END DEFINE ENVELOP
152. LOAD LIST 1 TO 11 21 TO 25
153. *PRINT JOINT DISPLACEMENTS ALL
154. LOAD LIST 26 TO 30 41 TO 44
155. *PRINT MEMBER FORCES ALL
156. LOAD LIST ALL
157. PARAMETER 1
158. CODE BS5950
159. SGR 0 ALL
160. MAIN 1 MEMB 1 TO 5 10 TO 25 30 TO 39
161. *TRACK 2
162. CHECK CODE ALL

STAAD.Pro CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.12_5950-1_2000

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					
1 ST	C250X90X9	PASS	BS-4.8.3.3.1	0.193	42
		0.67 C	0.60	5.39	-
2 ST	C250X90X9	PASS	BS-4.8.3.3.1	0.193	28
		0.69 C	0.60	5.39	-
3 ST	H250X250X9	PASS	ANNEX I.1	0.014	28
		14.07 C	0.10	3.50	-
4 ST	H250X250X9	PASS	ANNEX I.1	0.014	28
		4.60 C	0.10	3.42	-
5 ST	H200X200X8	PASS	ANNEX I.1	0.025	28
		0.01 C	0.01	3.53	-
6 ST	C150X75X6.5	PASS	BS-4.8.3.3.1	0.038	28
		3.41 C	0.04	0.10	-
7 ST	C150X75X6.5	PASS	BS-4.8.3.3.1	0.032	28
		3.72 C	0.04	0.10	-
8 ST	C150X75X6.5	PASS	BS-4.8.3.3.1	0.024	42
		3.05 C	0.04	0.01	-
9 ST	C150X75X6.5	PASS	BS-4.8.3.3.1	0.033	28
		2.62 C	0.04	0.10	-
10 ST	H200X200X8	PASS	BS-4.2.3-(Y)	0.010	26
		2.78 C	0.00	0.00	0.00
11 ST	H200X200X8	PASS	BS-4.2.3-(Y)	0.010	26
		2.78 C	0.00	0.00	0.00
12 ST	H250X250X9	PASS	BS-4.8.3.3.2	0.018	42
		16.35 C	0.10	3.61	-
13 ST	H250X250X9	PASS	BS-4.8.3.3.2	0.019	22
		13.22 C	0.01	3.01	-
14 ST	H200X200X8	PASS	ANNEX I.1	0.025	28
		0.01 C	0.01	3.65	-
15 ST	C250X90X9	PASS	BS-4.8.3.3.1	0.242	42
		0.04 C	0.69	6.40	-
16 ST	C250X90X9	PASS	BS-4.8.3.3.1	0.242	28
		0.04 C	0.69	6.40	-
17 ST	C250X90X9	PASS	BS-4.8.3.3.1	0.005	41
		0.01 C	0.05	0.07	-
18 ST	C250X90X9	PASS	BS-4.8.3.3.1	0.148	28
		2.25 C	0.79	7.44	-
19 ST	C250X90X9	PASS	BS-4.8.3.3.1	0.148	42
		2.25 C	0.79	7.44	-
20 ST	H250X250X9	PASS	ANNEX I.1	0.051	28
		30.19 C	2.54	7.75	-
21 ST	H250X250X9	PASS	ANNEX I.1	0.050	42
		29.71 C	2.46	7.75	-
22 ST	H200X200X8	PASS	BS-4.8.2.2	0.066	28
		0.02 T	-0.05	-9.38	1.20

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					
23 ST	H250X250X9	PASS	ANNEX I.1	0.076	28
		22.22 C	7.44	3.52	-
24 ST	H250X250X9	PASS	ANNEX I.1	0.076	42
		22.22 C	7.44	3.52	-
25 ST	H200X200X8	PASS	ANNEX I.1	0.036	28
		0.03 C	0.78	3.51	-
26 ST	C150X75X6.5	PASS	BS-4.8.3.3.1	0.114	28
		12.82 C	0.09	0.51	-
27 ST	C150X75X6.5	PASS	BS-4.8.3.3.1	0.022	4
		1.84 C	0.03	0.06	-
28 ST	C150X75X6.5	PASS	BS-4.8.3.3.1	0.116	42
		12.17 C	0.09	0.62	-
29 ST	C150X75X6.5	PASS	BS-4.8.2.2	0.020	28
		6.43 T	-0.04	0.20	0.00
30 ST	H200X200X8	PASS	BS-4.2.3-(Y)	0.002	26
		1.29 C	0.00	0.00	0.00
31 ST	H200X200X8	PASS	BS-4.2.3-(Y)	0.002	26
		1.29 C	0.00	0.00	0.00
32 ST	H200X200X8	PASS	BS-4.2.3-(Y)	0.014	26
		0.30 C	0.00	0.00	0.00
33 ST	H200X200X8	PASS	BS-4.2.3-(Y)	0.014	26
		0.30 C	0.00	0.00	0.00
34 ST	H250X250X9	PASS	ANNEX I.1	0.025	28
		22.62 C	0.66	5.12	-
35 ST	H250X250X9	PASS	ANNEX I.1	0.026	42
		22.14 C	0.74	5.12	-
36 ST	H200X200X8	PASS	BS-4.8.2.2	0.039	28
		0.02 T	0.00	-5.61	1.20
37 ST	H250X250X9	PASS	ANNEX I.1	0.014	28
		8.60 C	0.89	1.67	-
38 ST	H250X250X9	PASS	ANNEX I.1	0.014	42
		8.60 C	0.85	1.67	-
39 ST	H200X200X8	PASS	ANNEX I.1	0.012	28
		0.03 C	0.00	1.67	-

***** END OF TABULATED RESULT OF DESIGN *****

163. FINISH